Digital Chip Design Program

RTL Design

Mohammed alkhalifah

Mohammed alnaif

Task #01

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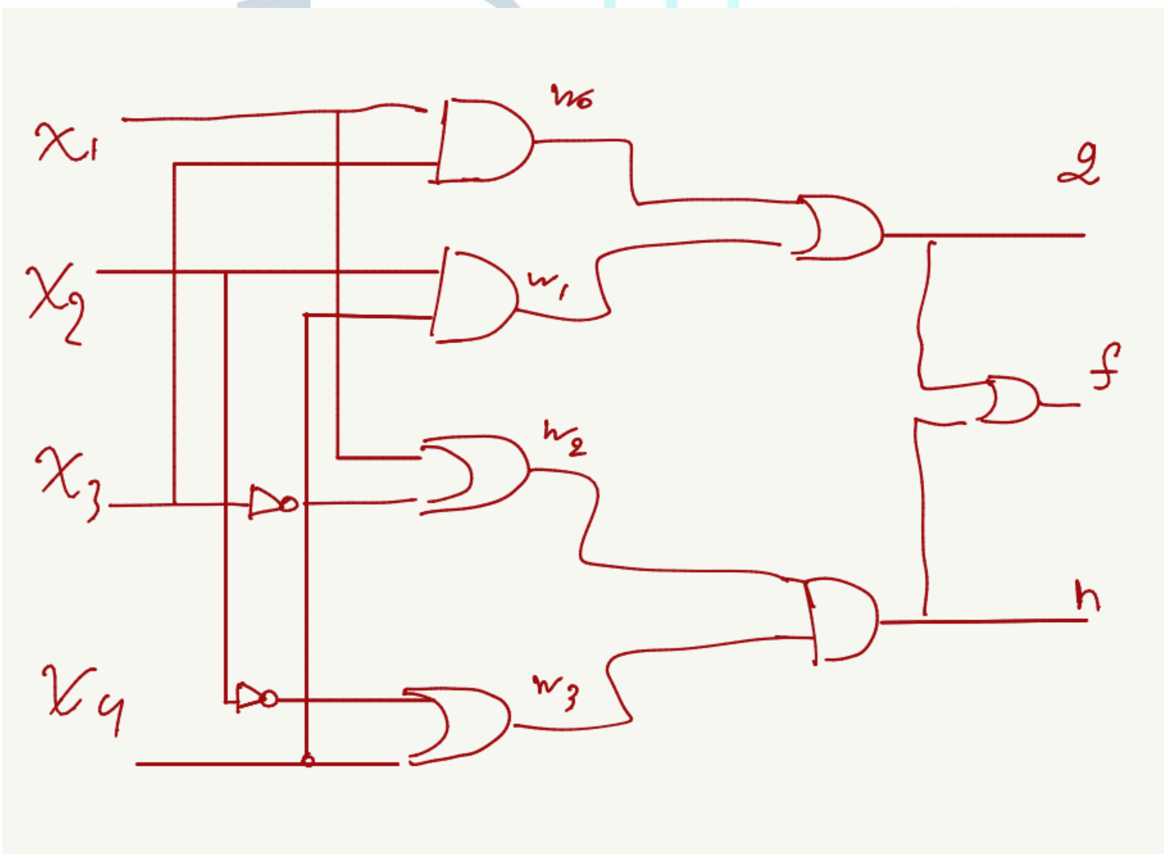
Gate-level

Task #1.1

```
module t1(x1,x2,x3,f);
   input  x1,x2,x3;
   output  f;
   wire [2:0]w;

   and (w[0],x1,x2);
   not(w[1],x2);
   and (w[2],x3,w[1]);
   or (f,w[0],w[2]);
endmodule
```

Task #1.2



Task #1.3

```
module t1(AeqB, AltB, AgtB, a, b);

input [3:0] a,b;
output AeqB, AltB, AgtB;
wire [3:0]i, w;

xnor (i[0], a[0], b[0]);
xnor (i[1], a[1], b[1]);
xnor (i[2], a[2], b[2]);
xnor (i[3], a[3], b[3]);
and (AeqB, i[0], i[1], i[2], i[3]);
and (w[0], a[3], ~b[3]);
and (w[1], a[2], ~b[2], i[3]);
and (w[2], a[1], ~b[1], i[3], i[2]);
and (w[3], a[0], ~b[0], i[3], i[2],i[1]);
or (AgtB, w[0], w[1], w[2], w[3]);
nor (AltB, AgtB, AeqB);

endmodule
```


Work distribution

Section	Abdulmalek	Omar	Khaled
Report	✓		
Task n.x		✓	
Task n.x+1			✓